Venkat Neelraj Nitta

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Summary

Software, data and ML engineering under one AI umbrella: 2+ years across e-commerce, finance and insurance, taking data from storage to decisions. I want to keep building on that as an AI Engineer.

Work Experience

- **Staples** Framingham, MA, US
  AI Engineer Intern *Jan 2026 – Jul 2026*
  - Designed the **evaluation** for a multi-turn LangGraph agent with function-calling tools and its product tagger: labeled sets plus an LLM judge; tagging accuracy averaged about 0.8, closer to 0.7 on buyer persona.
  - Maintained and extended the agent's **RAG** layer for agentic e-commerce: three Databricks Vector Search indexes and a Neo4j product graph, implementing its product-to-product relationships.
  - Built a fine-tuned multi-label DeBERTa tagger in PyTorch with an LLM fallback; moving batch inference over about 300K SKUs onto Ray on one GPU node cut runs from **about 4 hours to 90 minutes**.
  - Owned the daily pipeline feeding the agent: Databricks SQL merging more than 15 Snowflake tables into one Delta table, with validation, failure alerting, on-call support, a runbook and Azure DevOps CI/CD.
- **Cognida.ai** Hyderabad, India
  Software/AI Developer *Jul 2022 – Aug 2024*
  - Built embedding-based **RAG** for financial document search, cutting query resolution from 10 days to minutes; fine-tuned and quantized an LLM for an insurance-company chatbot to reduce serving compute cost.
  - Shipped unsupervised anomaly detection over millions of insurance claims (PySpark, topic modeling, K-means), flagging an additional 8% of new cases; ran distributed training on Ray and monitored GPU usage.
- **Lead Teaching Assistant**, Northeastern University, Boston, MA, US *Sep 2026 – Present*

Skills

GenAI & Agents: LangGraph, LangChain, RAG, LLM-as-Judge Evaluation, LLM Fine-Tuning, Hugging Face
ML & Data: PyTorch, Scikit-learn, Pandas, NumPy, SQL, PySpark, Databricks, Snowflake, Delta Lake, Ray, Neo4j
MLOps & Cloud: Docker, Kubernetes, MLflow, Azure DevOps, CI/CD, Databricks Model Serving, AWS, Azure
Programming: Python, TypeScript, FastAPI, Django, REST APIs, Unit & Integration Testing, Git, Linux
AI-Assisted Development: Claude Code, GitHub Copilot, Google Antigravity, Cursor, Codex

Projects

- **InsurAsst (FAISS, Kustomize, Gemini API) [GitHub]**: Claims assistant citing policy PDFs, with LLM coverage routing, a 12-scenario evaluation harness, per-turn traces and deployment manifests with health probes.
- **Pharma Agentic Capstone**: Binder-generation agents for a global pharmaceutical company, with evaluation; added guardrails to an HPC GPU-cluster agent: command allow-lists, prompt-injection and output checks, job limits.

Education

- **Northeastern University** Boston, MA, US
  Master of Professional Studies in Applied AI (STEM); GPA: 3.92 *Sep 2024 – Dec 2026 (Expected)*
  - Relevant Coursework: Fundamentals of Artificial Intelligence, AI System Technologies, Recommender System
- **Mahindra Ecole Centrale** Hyderabad, India
  Bachelor of Technology in Electrical and Electronics Engineering *Aug 2018 – Jun 2022*
  - Relevant Coursework: Machine Learning, Natural Language Processing, Data Structures, Statistics

Awards & Certifications

- **Hackathon**: Won the Workhuman-sponsored hackathon with Lighthouse (4-person team) [GitHub]
- **Certifications**: Generative AI with LLMs (DeepLearning.AI); AI Agents on Databricks (Databricks Academy)